

Supplementary material for “Designing European Hydrogen Market: Perspectives from Transmission System Operators”

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Abstract

Supplementary material for the publication “Designing European Hydrogen Market: Perspectives from Transmission System Operators”.

This document contains an introduction of concepts propedeutic for the main publication, as

Nomenclature

Indices and sets

$t \in \mathcal{T}$	Set of time steps t
$i \in \mathcal{P}$	Set of network users i
$u \in U$	Set of nodes u
$\mathcal{P}_u \subseteq \mathcal{P}$	Set of all shippers with a contract at node u

Variables

q_i	Supply or withdrawal of gas from player i [kg/h]
q_u	Total gas flow in node u [kg/h]
q_u^{TC}	Technical capacity at node u [MWh/h]
q_u^{FB}	Firm booked capacity at node u [MWh/h]
q_u^{FA}	Firm available capacity at node u [MWh/h]
q_u^{IT}	Interruptible capacity at node u [MWh/h]
$q_{i,u}^{\text{IB}}$	Interruptible capacity booked by shipper i at node u [MWh/h]
q_u^{IA}	Remaining available interruptible capacity at node u [MWh/h]
$q_{i,u}$	Capacity amount purchased by user i at node u [MWh/h]
$C_{i,u}$	Total capacity payment by user i at node u [€]

Parameters

λ_t	Day-ahead power price in time step t [€/MWh]
π_u^{book}	Booking price at node u [€/MWh]

1. EU Gas Market Principles in the EU

This section provides an extensive description of the foundational mechanisms of the EU gas market. Given the absence of comprehensive literature on the holistic functioning of gas markets, this section serves two purposes: centralising market operations and identifying the key market design components for gaseous commodities in the EU. The analysis covers market structure, capacity allocation, nomination procedures, balancing frameworks, and commodity trading to establish five core market design components that will guide the hydrogen market analysis.

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Figure 1: **The joint balancing zone of Denmark and Sweden.** Gas pipelines are shown in purple, storage points that serve as entry/exit points are in green, the virtual trading point is marked in orange, and cross-border entry/exit points are highlighted in yellow. Entry/exit points within the balancing zone are not displayed.

1.1. Entry-exit structure

The natural gas system is intended for the exchange of natural gas, achieved via a network of pipelines operated by the TSO. TSOs are responsible for the safe and reliable operation and maintenance of the gas network within a designated area defined as *balancing zone*. Notably, a single balancing zone can be managed by multiple TSOs; e.g., Germany’s balancing zone, which aligns with the national borders, includes several TSOs operating across the country. Historically, balancing zones have aligned with national borders. However, the recent introduction of joint balancing zones, spanning multiple national borders, has changed this. Examples include the Danish-Swedish joint balancing zone established in 2019, as well as the joint balancing zones between Belgium and Luxembourg and among Estonia, Latvia, and Finland. While the TSO is responsible for the high-pressure infrastructure, its operation is coupled to the Distribution System Operator (DSO), which operates the low-pressure distribution for end use and delivery to consumers.

The natural gas system is structured according to an *entry-exit system*, in which gas enters or exits the network at specific points designated by the TSO of the respective balancing zone. These points are classified as either entry points, where gas is injected into the system, or exit points, where gas is withdrawn, with some points serving both. Entry and exit points can be located within the balancing zone intended for internal gas exchange, or at its borders, facilitating cross-border exchange among neighbouring zones. Figure 1 illustrates the joint balancing zone of Denmark and Sweden, displaying only the cross-border entry/exit points, while omitting internal ones.

In the natural gas system, *network users*, also known as shippers, are parties who transport and exchange natural gas through the network. These include producers, who inject gas into the system; consumers, who withdraw gas at consumption points; and traders, who exchange gas within the market to leverage arbitrage opportunities.

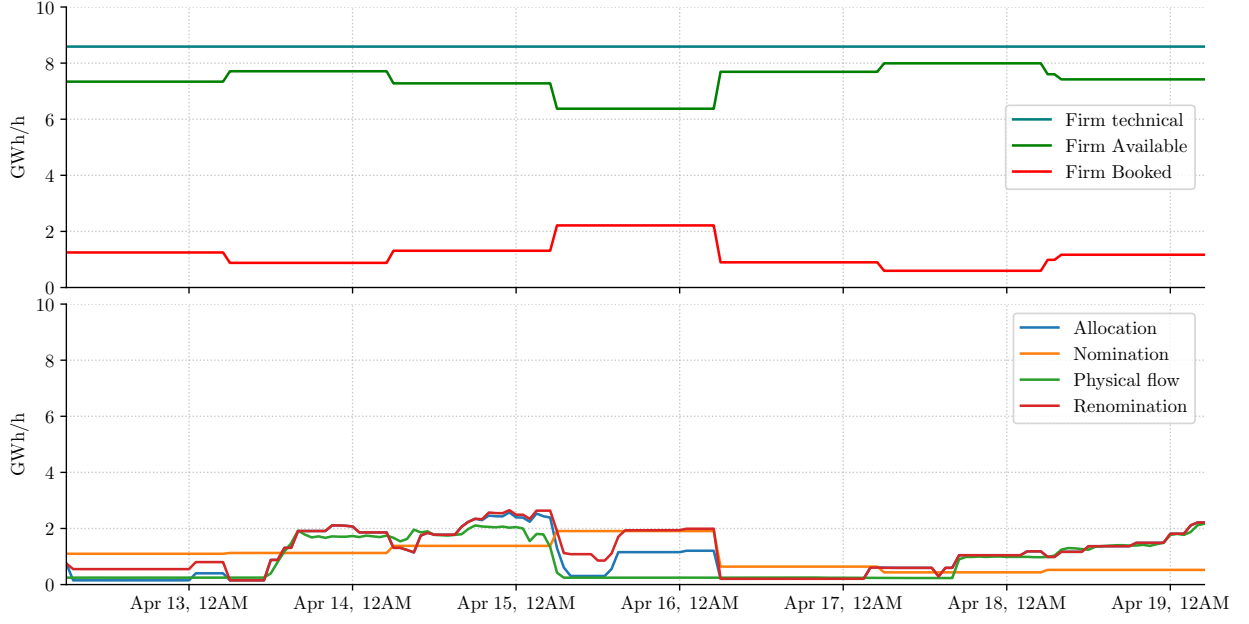


Figure 2: **Main figures for the Danish entry point of Nybro.** Data displayed from 12 April 2025 06:00 to 19 April 2025 05:00 in hourly resolution. The upper subplot shows the breakdown of firm capacity, whereas the lower subplot displays scheduling and allocation.

1.2. Capacity

Each entry and exit point u within the natural gas system set U is characterised by capacity q_u^{TC} , which is the maximum flow of natural gas that can be injected or withdrawn at that point, measured in MWh/h. Users must hold a capacity contract to obtain access rights for any network point, specifying the booked capacity and its duration. For example, a yearly capacity contract of 10 GWh/h at an exit point allows a user to withdraw up to 10 GWh in any given hour during that year. Holding capacity grants the right, but not the obligation, to flow gas at the specified point. Capacity products are available for purchase through online platforms, with PRISMA being the principal platform at the EU level (PRISMA, 2025). Notably, PRISMA is not the sole platform available; Poland and Hungary have also established official EU capacity platforms. These capacity products vary in duration (yearly, quarterly, monthly, daily, within-day) and type (firm or interruptible) and are regulated by the EU network codes (ENTSOG, 2017).

Firm capacity products guarantee uninterrupted access to the contracted volume for the entire duration of the agreement. Firm capacity contracts are auctioned on PRISMA. The total technical capacity at a network point u is auctioned, and the capacity purchased and booked by shipper i is defined as firm booked capacity $q_{u,i}^{\text{FB}}$. In contrast, the remaining unbooked portion is defined as the firm's available capacity, q_u^{FA} . For any point u , shipper $i \in \mathcal{P}$, these capacities are related by:

$$q_u^{\text{TC}} = q_u^{\text{FA}} + \sum_{i \in \mathcal{P}_u} q_{i,u}^{\text{FB}}, \quad \forall u \in U, i \in \mathcal{P}_u \quad (1)$$

where $\mathcal{P}_u \subseteq \mathcal{P}$ is the set of all shippers holding firm capacity contracts at point u .

In Figure 2, the upper plot illustrates historical capacity data for the Nybro entry point. The technical capacity remains constant at 8,589 GWh, while booked and available capacity fluctuate over time. The sum remains equal to the technical capacity.

Interruptible capacity is offered by TSOs to monetise or reschedule availability. These products grant conditional access subject to curtailment when firm capacity is fully utilised, allowing the firm to utilise other infrastructure while preserving the right to interrupt service during congestion.

Interruptible capacity is determined through three methods: first, as unused firm capacity; second, via counterflow-based assessment, where TSOs analyse historical flows in the opposite direction to estimate residual capacity; and lastly, based on statistical scenarios, which calculate the likelihood of interruption according to expected flow conditions within the system. The total interruptible capacity q_u^{IT} at node u is defined as the sum of interruptible capacity booked by shippers $q_{i,u}^{\text{IB}}$ and the remaining available interruptible capacity q_u^{IA} . These capacities are related as follows:

$$q_u^{\text{IT}} = q_u^{\text{IA}} + \sum_{i \in \mathcal{P}_u} q_{i,u}^{\text{IB}}, \quad \forall u \in U, i \in \mathcal{P}_u \quad (2)$$

For instance, if the technical capacity q_u^{TC} is 10 GWh/h, firm booked capacity q_u^{FB} is 10 GWh/h. Interruptible booked capacity q_u^{IB} is 5 GWh/h, firm users flowing 5 GWh and interruptible users flowing 5 GWh stay within system limits. However, if firm usage increases to 10 GWh, the TSO curtails interruptible flows to respect the technical capacity limit, leading shippers to interrupt their flows.

Capacity payment is defined as the product of the capacity booked by the shipper i at node u and its booking price π_u^{book} , applicable at each entry or exit node u . When a user i purchases a capacity amount $q_{i,u}$ at node u , the resulting capacity payment is calculated as follows:

$$C_{i,u} = q_{i,u} \pi_u^{\text{book}} \quad (3)$$

Capacity price, also referred to as capacity tariff, is regulated under the EU Tariff Codes (ENTSOG, 2018b), which establish general principles and set upper bounds for short-term pricing multipliers; individual TSOs determine the specific tariffs. To incentivise loyalty from shippers, TSOs commonly differentiate prices based on contract duration. For example, in Denmark and Germany, annual contracts are priced at the base rate π_u^{book} , whereas shorter-term contracts include premiums: quarterly at $1.1\pi_u^{\text{book}}$, monthly at $1.25\pi_u^{\text{book}}$, and daily at $1.4\pi_u^{\text{book}}$. These figures are not universally applied across the EU but are subject to national implementation within the limits defined by the EU framework. These capacity payments generally represent less than 10% of the total cost of gas delivery, with approximately 90% attributed to the cost of the gas itself, including taxes. Capacity payments constitute the primary revenue stream for TSOs, facilitating the operation and maintenance of the gas network. Revenue regulation is managed through the Regulatory Asset Base (RAB) framework, designed to ensure full cost recovery. If actual revenues exceed the allowable level (over-recovery), future tariffs are correspondingly reduced. Conversely, if revenues fall below the allowable level (under-recovery), tariffs are adjusted upward to compensate.

Capacity allocation is the competitive bidding process for access to capacity rights, which has historically transitioned from a first-come, first-served (FCFS) system to a regulated competitive bidding process under the EU's Third Energy Package (European Commission, 2009). The modern mechanism involves auctioning capacity products based on their duration: annual capacity is auctioned once per year, quarterly capacity is auctioned 1-4 times, monthly capacity is auctioned once per month, daily capacity is auctioned once per day, and within-day auctions occur multiple times a day. Auction formats vary by product type: daily and within-day auctions use uniform pricing, in which all successful bids pay the same clearing price, while for annual, quarterly, and monthly capacity products, an ascending-clock auction format is typically used. In the ascending clock auction, prices start at the base rate and increase incrementally over multiple rounds. This format allows participants to adjust their bids incrementally until the supply-and-demand equilibrium is reached.

1.3. Nomination, Renomination, allocation, and physical flow

Shippers submit their *nomination*, which is the schedule of hourly volumes (in MWh/h) they intend to inject or withdraw at designated entry and exit points after booking capacity. The

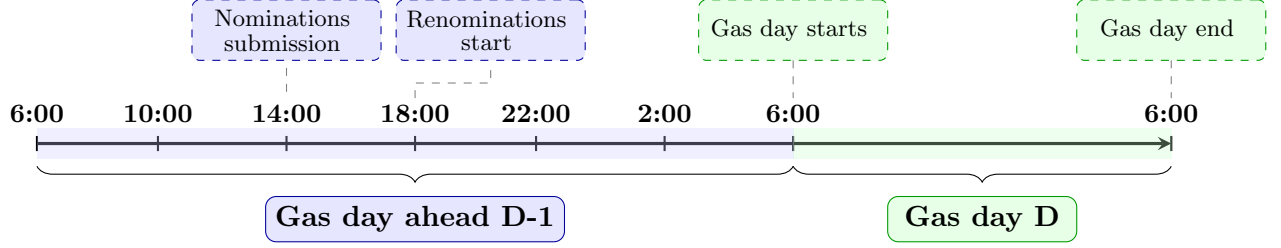


Figure 3: **EU Gas Day and Gas Day Ahead gate closures.** Schematic representation of the gas day-ahead market timeline showing key operational phases

European gas system operates on a standardised 06:00 to 06:00 CET as displayed in Figure 3, due to historical offtake patterns typically starting early in the morning. Nominations for gas day D must be submitted by 14:00 CET on D-1 and validated by TSOs between 14:00 and 18:00. Day-ahead and within-day capacity is booked after the first nomination rounds.

Renomination allows shippers to adjust their renominations to adapt to market conditions or system constraints. This process opens after 18:00 on D-1, allowing hourly adjustments with a two-hour lead time. For example, before 10:00 on gas day D-1, a renomination can be placed for all hours from 12:00 to 06:00, covering the remaining hours of the gas day. Renominations can cover the entire 24-hour gas day until 04:00 on gas day D. After 04:00, the renomination window shortens by one hour for each ensuing hour, with the final renomination window at 03:00 on gas day D, covering from 05:00 to 06:00.

Allocation is performed by the TSO at the end of each gas day. This accounting process assigns the actual metered gas volumes to individual shippers based on their nominations and the physical flow. If the physical flow matches the nomination, allocation equals nomination. However, if deviations occur due to renomination, network constraints, or metering errors, the renomination location reflects the reconciled volume actually delivered or received. Generally, at border points, allocation usually matches the nomination, while at consumption points, it is based on metering. Less frequent cases of modification to allocation accounting include the use of interruptible capacity or reductions in allocations due to unexpected maintenance.

Physical flow refers to the actual measured amount of gas transported and consumed, and it does not have to match the nomination or allocation exactly in real time. The system is optimised based on factors such as coordination with counterparts and operational constraints. Once nominations and renominations are in place, gas flows according to the updated schedules.

In Figure 2, the lower plot illustrates these dynamics. While the nominations are steady throughout the displayed period, renominations differ from the original schedule. These adjustments are reflected in the blue line (allocation), which updates accordingly. The green line shows the physical flow, which may deviate slightly but generally follows the contractual path.

Portfolios are associated with each shipper and serve as formal documentation of their booked capacities at various points, nomination schedules, and resulting allocations across all points where they hold capacity rights. In an entry-exit system, TSOs monitor the net position of each shipper's portfolio, which is the sum of all nominations across the portfolio. For example, if a shipper nominates +10 MWh at entry point A and -10 MWh at exit point B, the positions at each point are +10 MWh and -10 MWh, respectively, while the net position of the entire portfolio remains 0 MWh. Shippers must ensure that the portfolio's net position is balanced, meaning that nominations at entry points must correspond to matching exit points on the same day (conditional on WDOs). If a supply constraint is imposed on the shipper, requiring a change from 10 MWh to 9 MWh at entry point A, the shipper must renominate the exit volume to 9 MWh at exit point B to maintain a balanced portfolio.

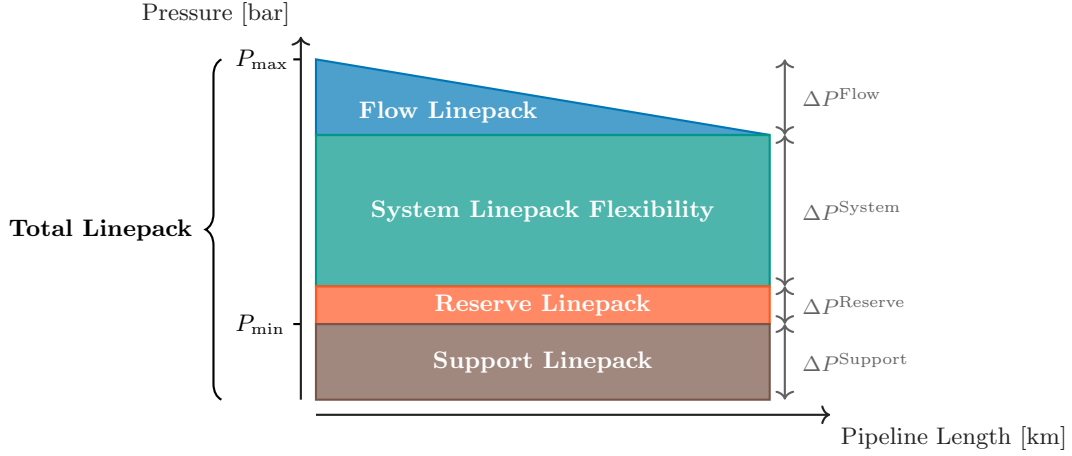


Figure 4: **Linepack flexibility regions in natural gas pipeline systems.** The diagram illustrates the pressure-dependent allocation of linepack capacity across different operational zones, from maximum system pressure ($P_{\max}$) to minimum operational pressure ($P_{\min}$). Illustration based on [Energinet \(2024\)](#).

1.4. Balancing framework

In an entry-exit system, the TSO manages the balance of its balancing zone by monitoring the net position of each shipper's portfolio. A portfolio's net position is long when it injects more gas than it withdraws (surplus) and short when it withdraws more gas than it injects (deficit) within a given timeframe. The *System Balancing Signal* (SBS) represents the aggregate net position of all portfolios within a balancing zone, reflects the overall system balance, and is therefore expressed in MWh. The SBS can be in three states: positive SBS indicates a long system with excess gas and higher-than-nominal pressure; negative SBS indicates a short system with a gas deficit and lower pressure; and zero SBS indicates a balanced system with no net imbalance.

Portfolios are classified based on their contribution to the system imbalance. Shippers whose net positions align with the direction of the imbalance are termed *causers*, while those whose net positions counteract the imbalance are termed *helpers*. When the system is long, long portfolios are causers and short portfolios are helpers, and conversely, when the system is short, short portfolios are causers and long portfolios are helpers.

From a physical perspective, a gas system is imbalanced when the pipeline pressure falls outside safety limits. To link the financial balancing of the portfolios to the physical pipeline balancing, the TSO introduced the concept of *linepack*. Linepack corresponds to the pressure of the gas stored in pipelines expressed in MWh and, according to Figure 4, is categorised into components: support linepack ensures minimum pressure levels, reserve linepack is allocated for TSO balancing actions and system survival, system linepack flexibility is available for flexible use by network users, and flow linepack enables hydrogen transport through pressure differences ([Energinet, 2024](#)).

The principles of the balancing scheme are outlined in the European Balancing Network Code, which regulates how TSOs should establish a balancing regime for their zones ([ENTSO, 2018a](#)). The end goal of a functioning balancing regime is to allow shippers to collectively balance their portfolios and minimise TSO minimisation through *balancing actions*, defined as interventions when the SBS moves outside predefined safety boundaries, requiring the TSO to buy or sell gas on the spot market. For systems exceeding limits, TSOs sell excess gas; for systems falling short, they buy gas. Since balancing actions are conducted at marginal spot market prices, they can be costly, with the costs distributed proportionally across the causer portfolios.

A common principle adopted among EU TSOs for evaluating the need for balancing action is the use of *buffer zones*, which divide the system linepack flexibility into zones. This is applied differently among EU countries, for example, the Dutch TSO Gasunie organises the following

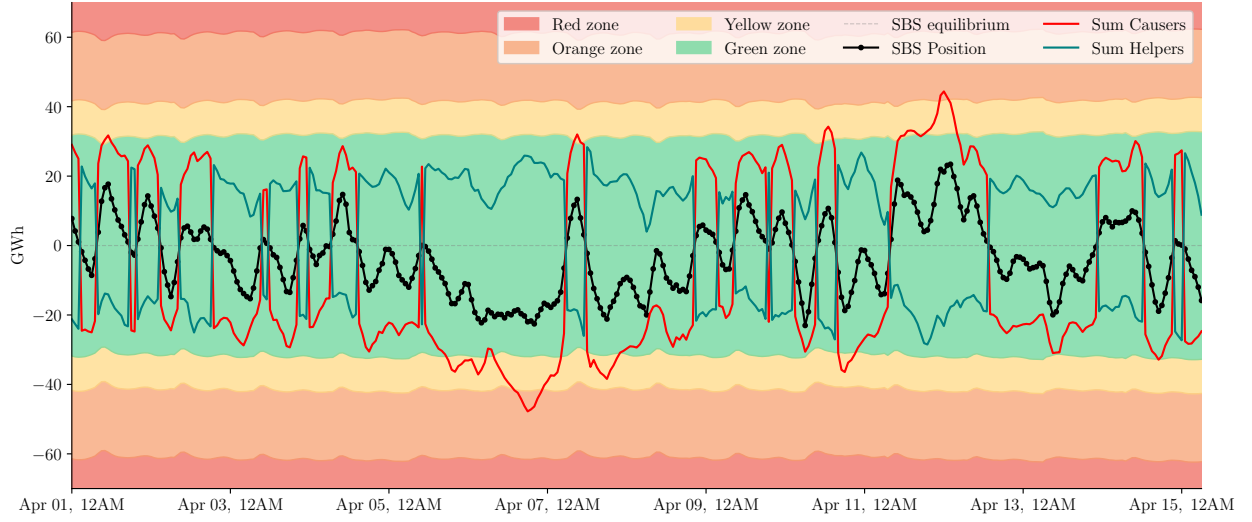


Figure 5: **Example of System Balancing Signal (SBS) and portfolio positions.** Data displayed for the Dutch balancing zone, from April 1th to 15th 2025 (Gasunie Transport Services (2025)).

(Gasunie Transport Services, 2025): the green zone operates with no balancing action as long as the SBS remains within limits; the yellow zone triggers end-of-day balancing actions under certain circumstances; the orange zone requires next-hour balancing actions when the SBS enters or remains in this zone; and the red zone demands immediate next-hour balancing actions.

Figure 5 illustrates the SBS evolution in the Dutch network over two weeks. On April 7th 2025, the SBS briefly changed from negative to positive and back again, temporarily reversing the roles of causers and helpers as the system fluctuated between long and short conditions. No balancing action was needed during this period because the SBS remained within the acceptable (green) zone.

Virtual Trading Points (VTPs) are non-physical entry-exit points within a balancing zone that allow gas to be traded between different market participants, such as producers and consumers, without requiring a physical gas exchange. VTPs are essential for enabling gas ownership transfers; without them, there is no basis for a market to function. Consider the example presented in Figure 6 of a balancing zone with shippers A and B: if Shipper A wants to sell gas to Shipper B, Shipper A injects 10 kWh of gas at an entry point, and Shipper B withdraws 10 kWh at an exit point, completing the physical gas exchange. Although the SBS of the zone is 0 kWh, individual portfolios are imbalanced as shown in the left plot of the figure: Shipper A's portfolio is long 10 kWh, and Shipper B's portfolio is short 10 kWh. To balance these portfolios without a VTP, Shipper A would need to purchase capacity at an exit point, physically exit and transfer the gas to Shipper B, who would then need to buy capacity at an entry point before reinjecting and withdrawing the gas. This process is operationally and financially inefficient due to additional capacity costs and the physical handling of the gas. The purpose of VTPs is to enable different entities (e.g., producers and consumers) to exchange gas.

In the right plot of Figure 6, the VTP allows shippers to transfer gas ownership virtually within the system, avoiding physical gas movement. As in the previous instance, Shipper A nominates the injection of gas, and Shipper B nominates the withdrawal. Additionally, both place a nomination at the VTP: Shipper A virtually exits the gas, and Shipper B virtually enters the gas, without any physical exchange. This virtual nomination balances the portfolios, ensuring the gas enters and exits the system only once. In reality, shippers submitting nominations to the VTP must specify trade amounts, delivery periods, and trade directions (entry or exit). For OTC trades, the TSO matches opposing nominations to complete the virtual transaction, whereas in exchange-based trades, matching is anonymous through the trading platform.

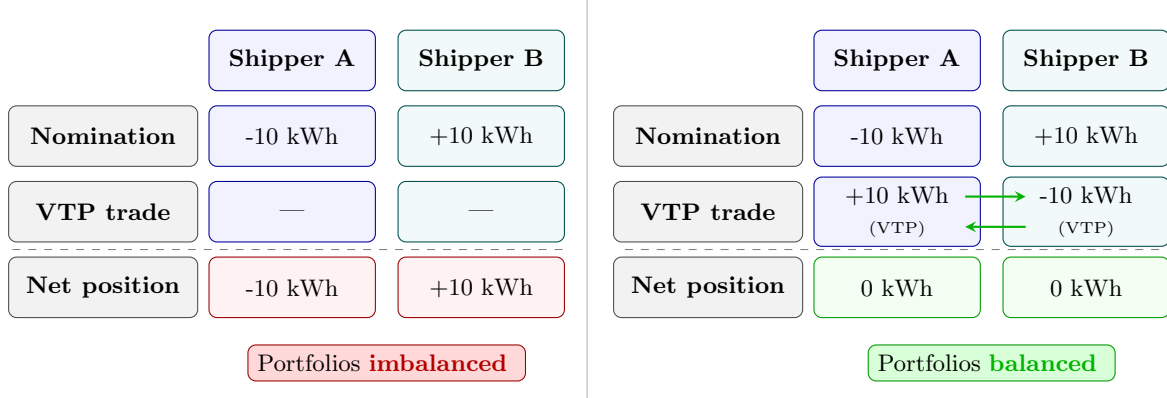


Figure 6: **Comparison of portfolio balancing mechanisms without (left plot) and with (right plot) Virtual Trading Points (VTP).** Without VTP, physical gas movement is required to balance portfolios; with VTP, gas ownership transfers are virtual, eliminating the need for physical movement and improving operational efficiency.

1.5. Commodity trading and gas spot market

Natural gas in Europe is traded through two main mechanisms: over-the-counter (OTC) bilateral contracts and centralised forms, both of which rely on virtual trading points to transfer gas ownership without physical movement. Before gas market liberalisation, companies were vertically integrated, supplying the commodity to the customers and managing transportation infrastructure. The EU Third Energy Package ([European Commission, 2009](#)) enforced unbundling, making TSOs responsible for infrastructure and allowing other parties to trade gas as a commodity.

OTC Contracts facilitate bilateral trades between buyers and sellers and are typically characterised by long-term duration and customisable schemes. Unlike standardised contracts, OTC contracts are fully customised by parties, allowing agreement on volume, duration, delivery profiles, and pricing structures. This flexibility is essential for long-term agreements (e.g., 30-year contracts) that standardised contracts cannot accommodate. Historically, OTC contracts have dominated gas markets because suppliers needed long-term agreements to secure bankability. However, because OTC contracts are bilateral and not traded on the market, they lack transparency in pricing and volumes traded, with external agencies reporting prices through broker and market participant surveys.

Exchange trading involves standardised platforms such as ICE (Intercontinental Exchange) and EEX (European Energy Exchange), which operate across both the US and European markets. These exchanges offer standardised contracts, including futures with delivery periods ranging from monthly to yearly. Unlike the electricity spot market, which clears hourly at a single price, the gas spot market operates as a continuous trading system with centralised participants who can execute transactions at any time during trading hours, resulting in continuous price points throughout the day. Exchanges calculate volume-weighted average prices at market close to determine the daily settlement price, commonly referred to as the "spot price." Initially, OTC contracts had the highest liquidity, but over time, exchanges have increased their liquidity.

The role of VTP is essential in ensuring the correct transfer of gas ownership between portfolios. Most gas trades, whether on exchanges or in OTC markets, use the VTP to transfer gas ownership among portfolios. For example, if a trade occurs between two shippers at EEX or ICE, the trade information is sent to the TSO, which inserts it into the shippers' portfolios via a nomination at the VTP. Exchange trades integrate automatically into nomination systems, while OTC transactions require manual or broker-supported nomination processes. VTP liquidity is the volume of trades executed at the VTP for that balancing zone. The largest VTP in Europe by liquidity is the Dutch Title Transfer Facility (TTF), followed by the British National Balancing Point (NBP). Although

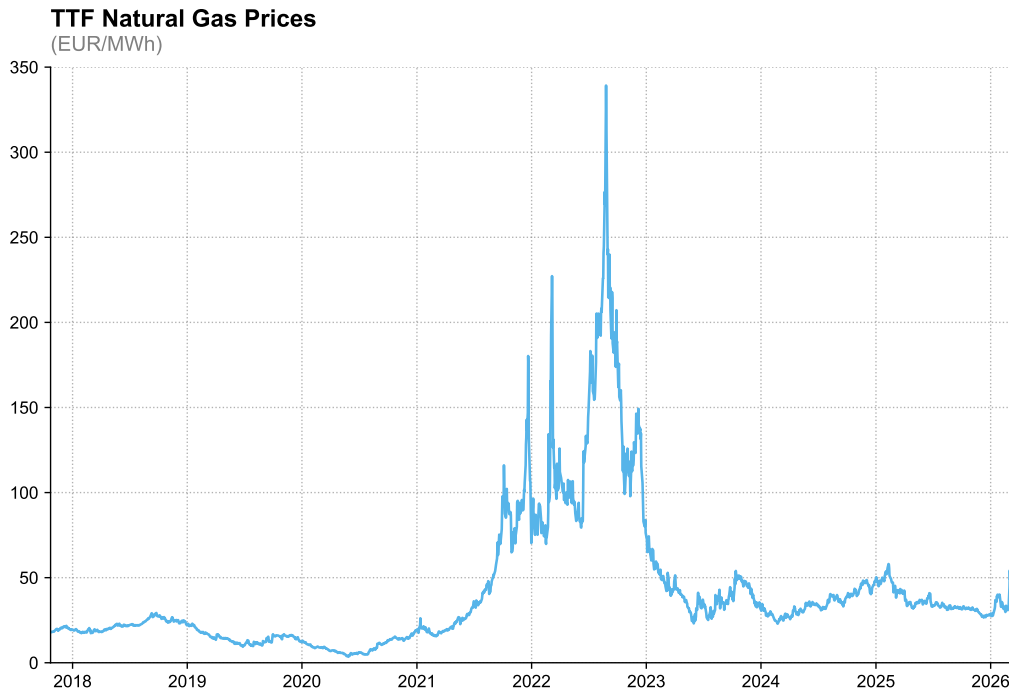


Figure 7: **Historical gas prices in the Dutch gas balancing zone.** Prices are referred to the TTF as proxy VTP in Europe (Yahoo Finance, 2025).

the term "VTP price" is commonly used to refer to the natural gas price, it is technically incorrect. The correct terminology is "EEX/ICE price of gas traded, settled at the TTF," although "TTF price" is widely accepted. The VTP price reflects volume-weighted averages of daily trades for specific products. As the most liquid VTP, the TTF serves as an effective continental price proxy, with its TTF price serving as the reference price, as seen in Figure 7. However, network congestion across balancing zones can lead to significant regional price deviations.

2. EU Energy Packages

The EU transformed its energy markets from monopolistic structures in the 1990s to competitive markets through five progressive Energy Packages:

1. **First Energy Package (1996-1998)** Initiated energy market liberalisation with third-party network access and unbundling of Transmission System Operators (TSOs), separating competitive activities (production/supply) from natural monopolies (transmission/distribution).
Legislation: Electricity Directive 96/92/EC (1996), Gas Directive 98/30/EC (1998)
2. **Second Energy Package (2003)** Advanced liberalisation by enabling consumer choice of energy suppliers, requiring legal unbundling of TSOs, and establishing independent National Regulatory Authorities (NRAs).
Legislation: Electricity Directive 2003/54/EC, Gas Directive 2003/55/EC, Regulation 1228/2003 (cross-border electricity access)
3. **Third Energy Package (2009)** Further enhanced TSO unbundling requirements, moving beyond legal separation to ensure complete independence of supply and generation activities

from network operations. This addressed the limitation where early market liberalisation only allowed trading at network entry and exit points, which hindered efficient commodity exchange.

The package strengthened NRA independence, established ACER, and created ENTSO-E and ENTSO-G for cross-border coordination. It introduced EU network codes, including Gas Balancing NC, Capacity Allocation Mechanisms NC, Congestion Management Procedures, and Harmonised Transmission Tariff Structures NC.

Key Innovation: VTPs as an administrative concept enabling gas commodity exchanges within the network system. These virtual points allow parties to trade gas while it remains in the system, facilitating the development of competitive exchanges and OTC contracts.

Legislation: Electricity Directive 2009/72/EC, Gas Directive 2009/73/EC, ACER Regulation 713/2009/EC, Electricity Regulation 714/2009/EC, Gas Regulation 715/2009/EC

4. **Fourth Energy Package / "Clean Energy Package" (2019)** Modernised the EU energy system with consumer empowerment tools and established 2030 climate targets: 40% GHG emission reduction, 32% renewables, 32.5% energy efficiency.

Legislation: Four directives (Electricity 2019/944, RED II 2018/2001, Energy Efficiency 2018/2002, Buildings Performance 2018/844) and four regulations (Electricity 2019/943, Risk Preparedness 2019/941, ACER 2019/942, Governance 2018/1999)

5. **Fifth Energy Package (2024)** Aligned energy policy with climate neutrality goals and extended the gas framework to hydrogen. Incorporated the REPowerEU plan to accelerate the Russian fossil fuel phase-out, enhance energy diversification, and reform electricity markets with stronger long-term contracts and consumer protections.

3. RFNBO Legislation

The Renewable Energy Directive (RED) establishes the EU's framework for the development of clean energy across all sectors, aiming to achieve climate neutrality by 2050. As part of its decarbonisation strategy, the EU aims to develop Renewable Fuels of Non-Biological Origin (RFNBO) such as renewable hydrogen, essential for decarbonising energy-intensive industrial processes and transport sectors where direct electrification is not feasible.

To produce RFNBO, strict criteria must be met to ensure that the electricity used is renewable. The Delegated Regulation (EU) 2023/1184 outlines detailed rules for determining when electricity can be considered fully renewable:

- **Electricity Sourced via Direct Connection:**

1. Electricity obtained directly from a renewable energy installation connected to the RFNBO production site qualifies as fully renewable without additional requirements.

- **Electricity Sourced from the Grid:** When the electricity for RFNBO production is sourced from the grid, several stringent conditions must be met:

1. **Additionality:** The renewable energy installation must have commenced operation not earlier than 36 months before the electrolyser and must not have received public financial support. This ensures that the renewable electricity is additional and not diverting existing resources.
2. **Geographical Correlation:** The renewable electricity source and the hydrogen production site must be located in the same bidding zone to avoid grid congestion and ensure efficient energy transfer.

3. **Temporal Correlation:** Until 2029, the electricity consumption and production must correlate within the same month to ensure the renewable electricity is actively contributing to hydrogen production. From 2030, this correlation must occur hourly, aligning more closely with real-time production and consumption.

The Delegated Regulation (EU) 2023/1185 complements these criteria by establishing a methodology for calculating the lifecycle greenhouse gas emissions associated with RFNBOs, ensuring comprehensive sustainability.

Despite these regulations, there are criticisms about the strictness of the criteria, which may limit the rapid deployment of renewable hydrogen production necessary for achieving the EU's climate goals. These concerns highlight the need for balancing stringent sustainability requirements with practical implementation to foster the growth of a viable hydrogen market.

4. Gas Congestion Management

Gas congestion occurs in two forms: commercial congestion, when capacity is booked but unutilized, and physical congestion, when infrastructure constraints prevent required gas flows. Commercial congestion can be addressed through reallocation mechanisms, while physical congestion requires infrastructure expansion.

Congestion typically manifests when significant price differences exist between gas hubs, creating arbitrage opportunities that drive shippers to book transport capacity rapidly. This phenomenon is particularly pronounced in short-term markets through day-ahead or within-day auctions. Physical constraints may exacerbate congestion during extreme weather events when supply limitations create security concerns. TSOs employ two categories of congestion management tools: direct or indirect.

Direct tools are specifically designed for congestion management:

1. **Use It or Lose It (UIOLI)** enables regulators to reallocate unused booked capacity. However, UIOLI implementation remains rare in Europe due to complex regulatory requirements involving multiple National Regulatory Authorities and legal concerns regarding capacity reallocation from private entities.

Indirect tools serve other primary purposes while alleviating congestion as a secondary effect:

1. **Auctions** create pricing mechanisms that impose premiums during congestion periods, making inefficient capacity use costly and ensuring fair allocation.
2. **Counterflow** optimises capacity utilisation by enabling commercial transactions in both directions at bidirectional connection points. While gas flows physically in only one direction at any given time, commercial capacity can be sold bidirectionally. This allows offsetting flows (e.g., 100 units north and 100 units south resulting in zero net physical flow) while maintaining commercial capacity utilization.
3. **Overnomination** provides short-term interruptible capacity that can be nominated up to two hours before the gas hour. Payment occurs only upon nomination, and this capacity receives the lowest priority in the allocation hierarchy: (1) Firm capacity, (2) Interruptible capacity, (3) Within-day interruptible capacity, (4) Overnomination. This mechanism maximizes available capacity utilization while providing flexibility for last-minute demand adjustments.

The combination of these tools, particularly auction-based allocation with appropriate pricing mechanisms, has proven effective in reducing congestion while maintaining efficient capacity utilization across European gas networks.

Data availability

Survey data will be made available on request. Supplementary material, figures and code are available at the [GitHub repository](#).

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